Genotype-by-Environment Interaction with Broiler Genotypes Differing in Growth Rate. 1. The Effects of High Ambient Temperature and Naked-Neck Genotype on Lines Differing in Genetic Background[1]

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ABSTRACT High ambient temperature (AT) significantly depresses growth rate and meat yield of commercial broilers, thus making it a major factor hindering poultry meat production, especially in hot climates. The effects of high AT were variably moderated when feather coverage was reduced by the naked-neck gene (*Na*). In this study, the effects of high AT and *Na* were investigated in broiler progeny of hens from a sire line and two dam lines, differing in growth rate and meat yield due to different breeding histories. Heterozygous naked-neck (*Na*/*na*) males were mated with normally feathered (*na*/*na*) hens from the three lines. The 500 progeny were segregated for *Na*/*na* and *na*/*na* genotypes. After brooding, chicks of each maternal background, *Na* genotype, and sex were equally divided to two similar chambers and were reared on litter to 53 d of age. One chamber was set to normal AT, averaging 25 C; the second chamber was set to high AT, averaging 30 C. The high AT treatment reduced growth and meat yield in the progeny of all three groups. This reduction increased with age and was highest in the broilers produced by hens from a sire line bred for high growth rate and breast meat yield. The two other groups, produced by hens from selected and relaxed dam lines, differed in growth rate but were similarly affected by the high AT. It is suggested that the magnitude of the high AT effect depends not only on differences in potential growth rate but also on differences in overall genetic background. It was also shown that broiler performance in the final weeks could be improved by introducing the *Na* gene into commercial flocks. The advantage of the *Na*/*na* genotype was much more pronounced at high AT and in broilers with genetically higher growth rate and breast meat yield.

(*Key words*: high ambient temperature, broiler, naked-neck gene, growth rate, breast meat)

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INTRODUCTION

Growth rate (GR) and meat yield of contemporary commercial broilers are significantly depressed by environmental stress caused by high ambient temperatures (AT) (Howlider and Rose, 1987, 1989; Osman et al., 1989; Cahaner and Leenstra, 1992; Leenstra and Cahaner, 1992; Geraert et al., 1996; Yahav et al., 1996; Yalcin et al., 1997a; Settar et al., 1999). Therefore, high AT has been a major factor hindering poultry meat production in hot climates, especially in developing countries where farmers cannot afford costly artificial control of AT in broiler houses.

Broiler lines with higher GR appear to be more sensitive to high AT. Pym and Farrell (1977) and Pym et al. (1984) found higher internal heat production in broiler lines selected for high GR and high feed consumption than in lines selected for other traits. Genetic heat tolerance, naturally occurring in a rural breed from a hot region (Arad and Marder, 1982) or developed by experimental selection (Bohren et al., 1982, 1983), was shown to be associated with low GR. Cahaner and Leenstra (1992) reported a larger reduction in actual GR due to constant high AT (32 C) in lines selected for high GR at normal AT than in lines with lower GR (due to selection for other traits). The latter study also suggested that carcass composition and meat yield might have been affecting heat tolerance in broilers.

However, broiler lines with different breeding histories may also differ in their sensitivities to high AT. Yalcin et al. (1997a) compared the performances of three commercial broiler lines developed by different breeding compa-

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Abbreviation Key: AT = ambient temperature; BR = breast meat weight; %BR = breast meat as percentage of live BW; GR = growth rate; HAT = high AT treatment (average: 30 C); NAT = normal AT treatment (average: 25 C); WG = BW gain (calculated average per day).

nies under natural hot (summer) and temperate (fall) climates in Turkey. The three lines exhibited similar GR in the temperate season but differed in the magnitude of the heat-induced reduction in their actual GR, leading to a significant line-by-season interaction. These results suggest that commercial broiler lines differ in their response to high AT, possibly due to differences in various aspects of their overall genetic background, such as original breeds, selection goals, and genetic drift.

Heat tolerance of fast-growing broilers can be improved by the naked-neck (*Na*) gene (Merat, 1986, 1990), which reduces feather coverage by about 15 to 20% in heterozygous (*Na*/*na*) and by 30 to 40% in homozygous (*Na*/*Na*) chickens (Cahaner et al., 1993; Yunis and Cahaner, 1999). Compared to their normally feathered counterparts, naked-neck broilers have a higher rate of heat dissipation (Yahav et al., 1998) and better thermoregulation in the heat (Deeb and Cahaner, 1999), resulting in higher actual GR and meat production at high AT (Cahaner et al., 1993; Yalcin et al., 1997b). In a more recent study, however, the effect of the *Na*/*na* genotype on performance at high AT differed in magnitude between commercial and noncommercial broiler lines, possibly due to a large difference in potential GR or due to different breeding histories (Yunis and Cahaner, 1999).

The objective of this study was to compare the effects of high AT and the naked-neck gene on actual GR of three broiler genetic backgrounds differing in their potential GR and meat yields due to breeding history: sire line vs. dam lines and GR selected vs. GR relaxed.

MATERIALS AND METHODS

Parental Lines and Crosses

Ten heterozygous naked-neck (*Na*/*na*) males from a relatively fast-growing commercial sire line (Cahaner et al., 1993) were mated with normally feathered hens from three different lines: G1, a commercial sire line continuously selected for high GR and breast meat yield; G2, a commercial dam line selected also for reproductive performance, hence with less emphasis on GR and breast yield; and G3, a noncommercial line derived from G2 and kept for five generations without selection for GR and meat yield. Each of the 10 males was mated with three to five hens from each of these three lines. A total of 500 chicks (about 170 from each of the three maternal backgrounds) were produced in a single hatch, with equal numbers of naked-neck (*Na*/*na*) and normally (*na*/*na*) feathered chicks. At hatch, all chicks were vent-sexed and wing-banded; they were reared on wood shavings in a single pen to 3 wk of age.

AT and Husbandry

On Day 1, AT was set to 35 C, then gradually reduced by 2 C every 3 d, to 23 C on Day 21. After this 3-wk brooding period, chicks of each maternal background, *Na* genotype, and sex were equally divided, placed in two similar chambers, and reared on litter up to 53 d of age. One chamber was set to a normal AT treatment (designated NAT), averaging 25 C with daily fluctuations between 20 and 30 C. The second chamber was set to a high AT treatment (designated HAT), averaging 30 C with daily fluctuations between 26 and 33 C.

Measurements

Each chick was weighed on Days 21, 28, and 42 and finally on Day 53, when the experiment was terminated. Average daily weight gain (WG) was calculated for each age interval. About 100 males (6 to 10 males from each combination of *Na* genotype, maternal background, and AT treatment) were processed on Day 53. Breast meat was dissected from each bird, and breast yields were recorded as absolute weight (BR) and as a percentage (%BR) of live BW.

Statistical Analysis

The data were subjected to a four-way ANOVA according to the model

$$Y = \mu + AT + NA + MB + S + AT \times NA + AT \times MB + AT \times S + NA \times MB + NA \times S + MB \times S + AT \times NA \times MB + e$$

with the main effects of AT, *Na* genotype (NA), maternal background (MB), and sex (S). There are six two-way interactions and a three-way interaction between AT, *Na* genotype, and maternal background; and e represents the random error. The three-way interactions with sex and the four-way interaction were not included in the final model because they were of nonsignificant and negligible magnitude in all traits.

Because only males were slaughtered, BR and %BR data were analyzed by a three-way model similar to the previous one except for the main effect of sex and the interactions with it

$$Y = \mu + AT + NA + MB + AT \times NA + AT \times MB + NA \times MB + AT \times NA \times MB + e.$$

Differences between least-square means and interactions between subsets of factor combinations were tested by contrasts by using a *t*-test. All statistical analyses were conducted by JMP® (SAS Institute, 2000).

RESULTS

As expected, sex had a highly significant effect on BW and WG but did not interact significantly with other factors; the *P*(F) values of these interactions are therefore not presented. Although no three-way interactions between AT, maternal background, and *Na* genotype were significant, detailed means (over sexes) of each combination of these three factors are given in Table 1 (for BW), Table 2 (for WG), and Table 3 (for BR and %BR). Several

TABLE 1. Mean BW at 21, 28, 42, and 53 d of age of normally feathered (*na/na*) and naked neck (*Na/na*) broilers (males + females) from three maternal background groups,[1] reared at normal (NAT[2]) or high (HAT[3]) ambient temperatures, and significance levels of the effects and differences

Variable	Broiler group[1]	*na/na*		*Na/na*		*Na* averages		
		NAT[2]	HAT[3]	NAT	HAT	NAT	HAT	$\%D_1$[4]
BW21	G1	594	595	617	584	606[a]	590[a]	–2.6
	G2	546	551	574	566	560[b]	559[b]	–0.2
	G3	516	512	526	501	521[c]	507[c]	–2.7
Group averages		552	553	572	550			
$\%D_2$[5]		0.2		–3.8				
BW28	G1	944	923	985	922	965[a]	923[a]	–4.4
	G2	861	855	897	898	879[b]	877[b]	–0.2
	G3	808	797	313	794	811[c]	796[c]	–1.8
Group averages		871	858	898	871			
$\%D_2$[5]		–1.5		–3.0				
BW42	G1	1,807	1,652	1,917	1,767	1,862[a]	1,710[a]	–8.2***
	G2	1,637	1,551	1,790	1,752	1,714[b]	1,652[b]	–3.6**
	G3	1,619	1,483	1,644	1,578	1,632[c]	1,531[c]	–6.2***
Group averages		1,688	1,562	1,784	1,699			
$\%D_2$[5]		–7.5***		–4.8***				
BW53	G1	2,535	1,970	2,588	2,271	2,562[a]	2,121[a]	–17.2***
	G2	2,235	1,986	2,416	2,262	2,326[b]	2,124[a]	–8.7***
	G3	2,197	1,866	2,271	2,102	2,234[b]	1,984[b]	–11.2***
Group averages		2,322	1,941	2,425	2,212			
$\%D_2$[5]		–16.4***		–8.8***				

ANOVA significance levels [*P*(F)][6]

	AT	NA	MB	AT × NA	AT × MB	NA × MB	AT × NA × MB
BW21	NS	NS	<0.001	NS	NS	NS	NS
BW28	0.058	0.063	<0.001	NS	NS	NS	NS
BW42	<0.001	<0.001	<0.001	NS	0.045	0.008	NS
BW53	<0.001	<0.001	<0.001	<0.001	<0.001	NS	NS

[a–c]Group means (in columns) with no common superscript differ significntly.

[1]Groups represent three maternal backgrounds. They were produced by the same sire line males and by hens from a commercial sire line (G1), a commercial dam line (G2), and a line derived from G2 with no BW selection for five generations.

[2]NAT = normal ambient temperature, averaging 25 C.

[3]HAT = high ambient temperature, averaging 30 C.

[4]$\%D_1$ = differences (in percentage) between AT means: 100 × (HAT – NAT)/NAT.

[5]$\%D_2$ = differences (in percentage) between AT means within *Na* genotypes: 100 × (HAT – NAT)/NAT.

[6]Model abbreviations: AT = ambient temperature; NA = *Na* genotype; MB = maternal background.

,*AT means (HAT – NAT) differ significantly: *P*(t) < 0.01 or 0.001, respectively.

significant two-way interactions were found between AT and maternal background and between AT and *Na* genotype; hence, means for these combinations are also presented (Tables 1, 2, and 3).

Maternal Backgrounds

As expected, maternal background significantly affected chick BW as early as 21 d of age. Mean BW of the lines at NAT ranked from G1 (highest) to G3 (lowest) at all ages (Table 1). Similarly, WG at NAT of the three groups also ranked from G1 to G3, although the genetically related G2 and G3 groups only differed significantly in their mean WG to Day 28 and not thereafter (Table 2).

Relative to NAT, HAT conditions depressed broiler BW and WG in all three groups. This effect increased with BW and WG due to differences in genetic background or age. The difference in mean WG between broilers at NAT and HAT was highly significant after Day 28 (i.e., WG28-42 and WG42-53, Table 2); consequently, they differed correspondingly in mean BW42 and BW53 (Table 1). However, the magnitude of the difference between NAT and HAT was much larger in the fastest-growing G1 group than in G2 and G3 groups, leading to a highly significant AT-by-maternal background interaction (AT × MB) for WG42-53 (Table 2) and BW53 (Table 1). Testing of contrasts between group means (data not shown) proved that these interactions were due to a larger NAT-HAT difference in the G1 birds, whereas the G2 and G3 groups were similarly affected by AT. The largest AT effects, and most significant AT × MB interaction, were observed for the last growing phase: WG42-53 of G1 birds was 41.2% lower at HAT than at NAT, whereas the corresponding differences in the G2 and G3 groups were 22.7 and 24.6%, respectively (Table 2). A similar interaction was observed for final weight: BW53 of G1 birds was 17.2% lower at HAT than at NAT and only 8.7 and 11.2% lower in G2 and G3, respectively (Table 1). Higher sensi-

tivity to heat in G1 vs. G2 or G3 was also apparent at earlier ages (Table 2). Means of WG21-28 and BW28 of birds at HAT and NAT were similar within the G2 and G3 groups, whereas a difference of 7.4% was observed in WG21-28 of G1 birds, leading to a –4.4% difference in BW28 of G1 birds at HAT vs. NAT. At WG28-42 and BW42, higher HAT-NAT differences were also observed in G1 birds vs. their counterparts in the G2 and G3 groups, although the contrasts between G1 vs. G2 and G3 were not large enough to produce a significant interaction.

At NAT, the dam line-derived G2 and G3 groups exhibited very similar breast meat yields in absolute weight (BR) and as percentage of BW53 (%BR), whereas significantly higher yield was exhibited by the sire line-derived G1 group (Table 3). The HAT-NAT differences in breast meat yield in the G1 group (–23.6 in BR and –5.8% in %BR) were significantly larger than the corresponding differences in the G2 and G3 groups (about 12% in BR and close to zero in %BR), leading to a significant AT × MB interaction in these two traits (Table 3).

Naked-Neck Genotypes

Because the AT treatments were initiated on Day 21, the similar mean BW21 of the normally feathered (*na/na*) and naked-neck (*Na/na*) birds within each group (Table 1) was expected. The naked-neck birds exhibited a higher mean WG21-28 at both NAT and HAT, although this effect, also observed also in BW28, was not significant. Mean WG28-42 and mean WG42-53 of the naked-neck birds were higher in all three groups in both AT environments, but the magnitude of the naked-neck advantage was larger at HAT than at NAT, leading to highly significant AT-by-Na genotype (AT × NA) interactions (Table 2). The reduction in WG28-42 due to HAT (relative to NAT) was 6.3% in the naked-neck broilers and more than twice that (13.9%) in the normally feathered broilers. A twofold heat effect in normally feathered vs. naked-neck broilers was also observed for WG42-53; the corresponding HAT-NAT differences were –40.1 vs. –20.1%, respectively (Table 2). Consequently, the *Na/na* genotype reduced the HAT vs. NAT difference in BW53 by half, from 16.4 to 8.8% (Table 1).

The naked-neck broilers performed better than their normally feathered counterparts in all three groups, as well as at NAT and HAT; hence, there was no significant AT × NA × MB interaction. A significant two-way interaction between the *Na* genotypes and maternal background (NA × MB) was found for WG28-42 and BW42 only,

TABLE 2. Means BW gain (WG) at three age intervals (Days 21 to 28, 28 to 42, and 42 to 53) of normally feathered (*na/na*) and naked-neck (*Na/na*) broilers (males + females) from three maternal background groups,[1] reared at normal (NAT[2]) or high (HAT[3]) ambient temperatures, and significance levels of the effects and differences

Variable	Broiler group[1]	*na/na*		*Na/na*		*Na* averages		
		NAT[2]	HAT[3]	NAT	HAT	NAT	HAT	$\%D_1$[4]
WG21–28	G1	50.1	46.9	52.6	48.2	51.4[a]	47.6[a]	–7.4
	G2	44.9	43.4	46.2	47.3	45.6[b]	45.4[b]	–0.4
	G3	41.8	40.7	40.9	41.8	41.4[c]	41.3[c]	–0.2
Group averages		45.6	43.7	46.6	45.8			
$\%D_2$[5]		–4.2		–1.7				
WG28–42	G1	61.6	52.0	66.5	60.4	64.1[a]	56.2[a]	–12.3***
	G2	55.5	49.7	63.8	61.1	59.7[b]	55.4[a]	–7.2***
	G3	57.9	49.0	59.4	56.0	58.7[b]	52.5[b]	–10.6***
Group averages		58.3	50.2	63.2	59.2			
$\%D_2$[5]		–13.9***		–6.3***				
WG42–53	G1	66.1	28.9	61.0	45.8	63.6[a]	37.4[b]	–41.2***
	G2	54.3	39.6	56.8	46.3	55.6[b]	43.0[a]	–22.7***
	G3	52.5	34.9	57.0	47.7	54.8[b]	41.3[a]	–24.6***
Group averages		57.6	34.5	58.3	46.6			
$\%D_2$[5]		–40.1***		–20.1***				

	ANOVA significance levels [*P*(F)][6]						
	AT	NA	MB	AT × NA	AT × MB	NA × MB	AT × NA × MB
WG 21–28	NS	NS	<0.001	NS	NS	NS	NS
WG 28–42	<0.001	<0.001	<0.001	0.006	0.030	0.007	NS
WG 42–53	<0.001	<0.001	NS	<0.001	0.002	NS	NS

[a–c]Group means (in columns) with no common superscript differ significantly.

[1]Groups represent three maternal backgrounds. They were produced by the same sire line males and by hens from a commercial sire line (G1), a commercial dam line (G2), and a line derived from G2 with no BW selection for five generations.

[2]NAT = normal ambient temperature, averaging 25 C.

[3]HAT = high ambient temperature, averaging 30 C.

[4]$\%D_1$ = differences (in percentage) between AT means: 100 × (HAT – NAT)/NAT.

[5]$\%D_2$ = differences (in percentage) between AT means within *Na* genotypes: 100 × (HAT – NAT)/NAT.

[6]Model abbreviations: AT = ambient temperature; NA = *Na* genotype; MB = maternal background.

***AT means (HAT – NAT) differ significantly: $P(t) < 0.001$.

TABLE 3. Mean breast meat weight (BW) and breast percentage (%BR) of normally feathered (*na/na*) and naked neck (*Na/na*) broilers (males only) from three maternal background groups,[1] reared at normal (NAT[2]) or high (HAT[3]) ambient temperatures, and significance levels of the effects and differences

Variable	Broiler group[1]	*na/na*		*Na/na*		*Na* averages		
		NAT[2]	HAT[3]	NAT	HAT	NAT	HAT	$\%D_1$[4]
		(g per bird)						
BR	G1	460	350	524	401	492[a]	376[a]	–23.6***
	G2	405	321	428	407	417[b]	364[b]	–12.7***
	G3	395	318	399	383	397[b]	351[a]	–11.6**
Group averages		420	330	450	397			
$\%D_2$[5]		21.4***		–11.8***				
		(% of BW)						
%BR	G1	17.1	15.3	17.4	17.2	17.3[a]	16.3[a]	–5.8*
	G2	15.7	15.0	15.5	16.4	15.6[b]	15.7[b]	0.6
	G3	15.7	14.9	15.8	16.5	15.8[b]	15.7[b]	–0.6
Group averages		16.2	15.1	16.2	16.7			
$\%D_2$[5]		–6.8**		+3.1				

	ANOVA significance levels [*P*(F)][6]						
	AT	NA	MB	AT × NA	AT × MB	NA × MB	AT × NA × MB
BR	<0.001	<0.001	<0.001	<0.061	<0.005	NS	NS
%BR	NS	0.002	<0.001	<0.009	<0.045	NS	NS

[a–c]Group means (in columns) with no common superscript differ significntly.

[1]Groups represent three maternal backgrounds. They were produced by the same sire line males and by hens from a commercial sire line (G1), a commercial dam line (G2), and a line derived from G2 with no BW selection for five generations.

[2]NAT = normal ambient temperature, averaging 25 C.

[3]HAT = high ambient temperature, averaging 30 C.

[4]$\%D_1$ = differences (in percentage) between AT means: 100 × (HAT – NAT)/NAT.

[5]$\%D_2$ = differences (in percentage) between AT means within *Na* genotypes: 100 × (HAT – NAT)/NAT.

[6]Model abbreviations: AT = ambient temperature; NA = *Na* genotype; MB = maternal background.

*,**,***AT means (HAT – NAT) differ significantly: $P(t) < 0.05$, 0.01, or 0.001, respectively.

because the advantage of the naked-neck birds was larger in group G2 than in the other two groups. It should be noted, however, that WG after Day 28 and, consequently, final BW (BW53) were higher in the naked-neck than in the normally feathered birds in all maternal backgrounds. Averaged over the three groups, the advantage of the naked-neck birds was significant in both AT environments, but it was threefold greater at HAT (2,212 vs. 1,941 g, a 14% difference) than at NAT (2,425 vs. 2,322 g, a 4.4% difference) (Table 1).

The *Na/na* genotype increased BR in all three maternal backgrounds and in both AT environments, but the effect was greater at HAT than at NAT, reflecting the similar effects on WG and BW. Averaged over the three groups, mean BR at HAT was lower by 21.4% than that at NAT in normally feathered birds, whereas the corresponding difference in the naked-neck birds was only 11.8%, leading to a significant AT × NA interaction (Table 3). The reduction in %BR due to high AT was significant only in the normally feathered broilers (–6.8%) as compared to a nonsignificant increase of 3.1% in the naked-neck broilers.

DISCUSSION

The birds in all three groups were produced by the same sires, but the second half of each group's genetic background was contributed by one of three different maternal lines. The commercial sire line (G1) and dam line (G2) differed in their genetic potential for GR and breast meat yield. These differences were due to different histories (the G1 line was mainly based on the Cornish breed, and the G2 line was based on the White-Rock breed) and different selection emphases during many generations of selection. The G3 line was derived from G2; hence, dams of these two groups were expected to be genetically similar, except for a difference in BW at any given age, because the selection on GR had been relaxed in G3 for five generations. However, other genetic differences might have accumulated between G2 and G3 during these five generations: the G2 line had been through a large-scale commercial breeding program, whereas the G3 line was reproduced by only 40 sires and 200 dams per generation, hence the possibility of genetic drift cannot be excluded.

The actual performances of the three groups under NAT were in agreement with the known differences between the maternal backgrounds. Mean final BW of the G1 group was higher than that of the G2 group (2,562 vs. 2,326 g, respectively). These means and the corresponding differences in WG reflected about one-half of the known difference in GR between a sire line and a dam line. Only one-half of the difference was expected because the same sires were crossed with dams from the different lines. Mean final BW of the G3 group was about 100 g lower

than that of G2 (2,234 vs. 2,326 g, respectively), reflecting significantly lower WG up to 28 d of age but similar WG thereafter. Commercial breeding companies report an average genetic gain in BW of about 50 to 60 g per generation. Therefore, with a difference on the maternal side of five generations without selection on BW and using the same sires, the difference in BW between G2 and G3 was somewhat lower than the expected difference of about 140 g. Because the G2 line and the G2-derived G3 line were not selected for breast meat yield, their mean BR and %BR were indeed almost identical, whereas the G1 group exhibited significantly higher breast meat yield.

The magnitude of the negative effect of high AT on WG and BW increased with age and was larger in G1, the group with higher genetic potential for rapid GR. The WG of G1 broilers was reduced by HAT as early as Days 21 to 28 (–7.4%, Table 2), when no difference had yet been observed in the G2 and G3 groups. Growth during Days 28 to 42 (WG28-42) of the G1 group was 12.3% lower at HAT than at NAT, whereas the corresponding reduction in G2 and G3 averaged only 8.9%. During the last 11 d, the relative difference in WG42-53 of the birds reared at HAT vs. NAT was –41.2% in G1 and only about –23% in G2 and G3. This case of genotype-by-environment interaction can be interpreted as an indication of higher sensitivity to high AT in broiler genotypes with a higher potential for rapid GR. A similar negative association between potential GR (WG in temperate climate) and sensitivity to high AT (actual WG in hot climate) was found in a recent genotype-by-environment study (Settar et al., 1999). If the magnitude of the reduction in WG caused by high AT increases as the genetic potential for high WG increases, commercial broilers would be expected to become increasingly sensitive to high AT because of their continuous selection for rapid growth at optimal AT.

However, despite the difference of about 100 g between mean BW of the G2 and G3 birds, both groups were similarly affected by the high AT. This result could be related to their nearly identical breast meat yield, which was significantly lower than that of the G1 group. Cahaner and Leenstra (1992) concluded that differences between lines in their growth at high AT are related to carcass composition, as well as to genetic potential for rapid GR but not to actual BW. At the normal AT, the G1 group, representing a commercial sire line heavily selected for rapid GR and high breast meat yield, averaged 492 g breast meat per bird, 85 g (21%) more than the average BR of G2 and G3. A high rate of protein synthesis in chickens was found to be associated with higher internal heat production (Muramatsu and Okumura, 1985; Muramatsu et al., 1987), leading to lower heat tolerance. Therefore, the similar heat tolerance of the G2 and G3 groups, which was higher than that of the G1 group, could be related to their similar breast meat yields. However, with only five generations apart, the G2 and G3 groups still have the same genetic background (except for a small difference in GR up to 28 d of age). The Cornish breed, the base population of current broiler sire lines (such as G1 in this study) was very different from the breeds used as base populations of broiler dam lines (e.g., G2 and G3). After many generations of separate breeding programs with different goals, the genetic background of the G1 line should be very different from that of G2 and G3. Therefore other traits, besides GR and breast meat yield, might have been responsible for the observed line differences (G1 vs. G2 + G3) in heat tolerance.

The three groups in this study were progeny of heterozygous naked-neck (*Na*/*na*) males and normally feathered females (*na*/*na*); hence, they consisted of an equal number of naked-neck and normally feathered broilers. At high AT, the naked-neck broilers were expected to exhibit higher WG and BW than their normally feathered counterparts because of their better thermoregulation (Deeb and Cahaner, 1999) and possibly also because of amino acids saved from feather production, hence becoming available for meat tissues (Cahaner et al., 1987). Yunis and Cahaner (1999) suggested that the superiority of naked-neck broilers over their normally feathered counterparts at high AT is greater in a line with higher genetic potential for rapid GR. The results from the present study support this hypothesis. In G1, the HAT-NAT difference in mean WG42-53 was –56% in the normally feathered birds, twice as great as the –25% difference between naked-neck G1 birds at HAT vs. NAT. In the G2 and G3 groups, the corresponding differences were about 30 and 17% for the normally feathered and naked-neck birds, respectively. Similar relationships were also observed in WG28-42, BW42, and BW53. However, the interactions between *Na* genotypes and maternal backgrounds were not significant because the differences between groups in the magnitude of the heat-induced GR reduction in normally feathered vs. naked-neck broilers were not large enough. In contrast to these results, a significant *Na* genotype-by-line interaction was found by Eberhart and Washburn (1993b). However, that study differed from our present study in three major respects: 1) The constant high AT (32 C), used by Eberhart and Washburn, has a greater negative effect on GR than fluctuating high AT (Yahav et al., 1996); 2) The average reduction in feather coverage of the naked-neck birds was more pronounced because homozygous (*Na*/*Na*) birds were also included and are known to be more tolerant to heat (e.g., Cahaner et al., 1993); 3) in their study, BW of the high-GR line (commercial broilers) was about twice that of the low GR line (the Athens-Canadian randombred population).

The last point limits the relevance of the study by Eberhart and Washburn (1993b) and similar studies. In order to assess the potential contribution of the *Na* gene to heat tolerance of present day and future commercial broilers, studies on naked-neck broilers should use relevant stocks. The continual change in the range of NAT used in broiler studies demonstrates the need for modern stocks. Naked-neck chickens exhibited lower WG and BW than their normally feathered counterparts at NAT (about 20 to 28 C) in studies during the 1980s (Merat, 1979, 1986; Hanzl and Somes, 1983; Zein el Dein et al., 1984). In contrast, the performance of the naked-neck birds at NAT in the present study was similar to or better than that of their

normally feathered counterparts. Several studies during the last decade have reported similar results (Cahaner et al., 1993; Eberhart and Washburn, 1993a,b; Yalcin et al., 1997b; Yahav et al., 1998; Deeb and Cahaner, 1999; Yunis and Cahaner, 1999). It appears that the AT level at which the reduced feather coverage turns from disadvantageous to beneficial is continually decreasing. The continuous selection for high GR has been increasing feed intake, metabolic rate, and internally generated heat per unit time (Emmans and Kyriazakis, 2000). Lower AT (below 20 C) are therefore needed for the modern, fast-growing, normally feathered broilers to efficiently dissipate the higher amount of internal heat (Yahav et al., 1996) and to achieve their potential growth rate (Emmans and Kyriazakis, 2000). Currently, already at AT levels of 20 to 25 C, naked-neck broilers exhibit an advantage, and this advantage increases at higher AT. As broilers continue to be selected for higher GR and meat yield (McKay et al., 2000), the AT for optimal broiler growth is expected to further decrease (Emmans and Kyriazakis, 2000), and, consequently, AT levels that are currently considered as normal are expected to substantially depress broiler performance. Under these circumstances, the advantage of naked-neck broilers over those normally feathered might be expected to increase.

In conclusion, the results of the present study demonstrate the drastic reduction in broiler performance due to the relatively small difference between normal (average 25 C) and high AT (average 30 C). The magnitude of the reduction in WG and BW appears to increase with higher GR and breast meat yield. Commercial breeding programs are continuously emphasizing these two traits, and hence the magnitude of the heat-induced depression in broiler performance might be expected to increase. This negative effect is partially alleviated by the naked-neck gene (*Na*), and the advantage of the naked-neck genotype is expected to increase in the future because of the higher dependence of fast-growing broilers on rate of heat dissipation. However, it is obvious that the *Na* gene alone cannot overcome broiler needs for lower AT. Therefore, it is important to confirm the association between high GR (and meat yield) and sensitivity to heat suggested by the results of the present study. It should be noted that in this study, as in previous studies (e.g., Arad and Marder, 1982; Bohren et al., 1982, 1983; Cahaner and Leenstra, 1992; Washburn et al., 1992; Eberhart and Washburn, 1993b; Yunis and Cahaner, 1999), the genetic groups differing in heat tolerance also differed in their overall genetic background, hence traits other than GR and %BR may be affecting their response to high AT. Therefore, the net effect of GR on heat tolerance needs to be determined by using birds that differ only in this trait but share the same genetic background.

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